

1. O/N/23/11/Q39

What is the first stage in the life cycle of stars?

- A** a black hole
- B** a cloud of gas and dust
- C** a red supergiant
- D** a supernova

2. O/N/23/11/Q40

What is the definition of redshift?

- A** the increase in observed frequency of light from galaxies that are moving away from the Earth
- B** the increase in observed wavelength of light from galaxies that are moving away from the Earth
- C** the light emitted by distant galaxies that is blue when it reaches the Earth
- D** the light reaching the Earth that is red when it is emitted by distant galaxies

3. O/N/23/12/Q40

When observed on the Earth, the redshift of the light from galaxy X is smaller than the redshift of the light from galaxy Y.

Which galaxy is closer to the Earth and which galaxy is receding from the Earth faster?

	closer to the Earth	faster recession
A	X	X
B	X	Y
C	Y	X
D	Y	Y

4. M/J/23/12/Q39

In which region of the electromagnetic spectrum does the Sun radiate the most energy?

- A** infrared region
- B** microwave region
- C** radio wave region
- D** X-ray region

5. M/J/23/12/Q40

Four of the stages in the life cycle of a star, until it becomes a red giant, are shown.

- W Inward force of gravitational attraction is balanced by an outward force from its centre.
- X Internal gravitational collapse produces an increase in temperature.
- Y It expands.
- Z Most of the hydrogen has been converted to helium.

In which order do these stages occur, starting with the earliest?

- A $W \rightarrow X \rightarrow Y \rightarrow Z$
- B $W \rightarrow X \rightarrow Z \rightarrow Y$
- C $X \rightarrow W \rightarrow Y \rightarrow Z$
- D $X \rightarrow W \rightarrow Z \rightarrow Y$

6. M/J/23/11/Q39

It takes 27 days for the Moon to orbit the Earth.

The average value of the radius of the Moon's orbit is 3.8×10^8 m.

What is the average orbital speed of the Moon?

- A 160 m/s
- B 1000 m/s
- C 1.4×10^8 m/s
- D 8.8×10^8 m/s

7. M/J/23/11/Q40

What is the nuclear reaction that takes place in stable stars?

- A fission of helium to produce hydrogen
- B fission of hydrogen to produce helium
- C fusion of helium into hydrogen
- D fusion of hydrogen into helium